ASSIGNMENT-9

① $m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = 0$

say $y_1 = x \rightarrow$ displacement

$y_2 = \dot{x} \rightarrow$ vel.

$$\frac{dy_1}{dt} = y_2$$

$$m\frac{dy_2}{dt} + cy_2 + ky_1 = 0$$

$$\frac{dy_2}{dt} = -\frac{k}{m}y_1 - \frac{c}{m}y_2$$

$$f_1(t, y_1, y_2) = \frac{dy_1}{dt} = y_2$$

$$f_2(t, y_1, y_2) = \frac{dy_2}{dt} = -y_1 - \frac{c}{20}y_2$$

② like if we want to move forward in time from

$t_i \rightarrow t_{i+1} \rightarrow$ step size (h)

$k_{i,j} = k_{ij}$

① $k_{11} = h \cdot f_1(t_i, y_{1i}, y_{2i})$

$k_{12} = h \cdot f_2(t_i, y_{1i}, y_{2i})$

$$k_{21} = h \cdot f_1\left(t_i + \frac{h}{2}, y_{1i} + \frac{k_{11}}{2}, y_{2i} + \frac{k_{12}}{2}\right)$$

$$k_{12} = h \cdot f_2\left(t_i + \frac{h}{2},\ y_{1i} + \frac{k_{11}}{2},\ y_{2i} + \frac{k_{12}}{2}\right)$$

$$k_{31} = h \cdot f_1\left(t_i + \frac{h}{2},\ y_{1i} + \frac{k_{21}}{2},\ y_{2i} + \frac{k_{22}}{2}\right)$$

$$k_{32} = h \cdot f_2\left(t_i + \frac{h}{2},\ y_{1i} + \frac{k_{21}}{2},\ y_{2i} + \frac{k_{22}}{2}\right)$$

$$k_{41} = h \cdot f_1\left(t_i + h,\ y_{1i} + k_{31},\ y_{2i} + k_{32}\right)$$

$$k_{42} = h \cdot f_2\left(t_i + h,\ y_{1i} + k_{31},\ y_{2i} + k_{32}\right)$$

→

$$y_{1,i+1} = y_{1i} + \frac{1}{6}\left(k_{11} + 2k_{21} + 2k_{31} + k_{41}\right)$$

$$y_{2,i+1} = y_{2i} + \frac{1}{6}\left(k_{12} + 2k_{22} + 2k_{32} + k_{42}\right)$$

③ $f(t,y) = \frac{dy}{dt} = -2y/t$

$h = 0.5$ $\qquad t: 2.5 \to 3 \to 3.5 \cdots$

$t_0 = 2.5,\ y_0 = 0.48 \Rightarrow f_0 = \frac{-2(0.48)}{2.5} = -0.384$

$t_1 = 3,\ y_1 = 0.333 \Rightarrow f_1 = -2\frac{(0.3332)}{3.0} = -0.2222$

$t_2 = 3.5,\ y_2 = 0.24499 \Rightarrow f_2 = -2(0.24499)/3.5 = -0.139943$

$t_3 = 4$, $y_2 = 0.1875 \Rightarrow f_3 = -2(0.1875)/4 = -0.09375$

$\Rightarrow y_4 @ t_4 = 4.5$ <u>AB method</u> 4th order

$$y_4^{(0)} = y_3 + \frac{h}{24}[55f_3 - 59f_2 + 37f_1 - 9f_0]$$

$$y_4^{0} = 0.1875 + \frac{0.5}{24}[55(-0.093750) - 59(-0.139943) + 37(-0.2222) - 9(-0.384)]$$

$$y_4^{(0)} = 0.1875 + (0.020833)[-5.15625 + 8.256687 - 8.2214 + 3.456]$$

$$y_4^{(0)} = 0.1875 + (0.020833)(-1.665013) = 0.152812$$

$$y_4^{(j)} = y_3 + \frac{h}{24}[9f(t_4, y_4^{(j-1)}) + 19f_3 - 5f_2 + f_1]$$

$$19f_3 - 5f_2 + f_1 = 19[-0.09375] - 5(-0.139943) + (-0.222)$$

$$= -1.78125 + 0.699715 - 0.222 = -1.303735$$

(i)

$$f_4^{(0)} = \frac{-2(0.152812)}{4.5} = -0.067916$$

$$y_4^{(1)} = 0.1875 + \frac{0.5}{24}[9(-0.067916)] - 1.303735]$$

$$y_4^{(1)} = 0.1875 + 0.020833[-1.914971] = 0.147605$$

$$\varepsilon_a = \left| \frac{0.147605 - 0.152812}{0.147605} \right| \times 100 = 3.528\%$$

it-2:-

$$f_y^{(1)} = -2(0.147605)/4.5 = -0.065602$$

$$y_y^{(2)} = 0.1875 + \frac{0.5}{4}[9(-0.065602) - 1.303735]$$

$$y_y^2 = 0.1875 + 0.020833[-1.894153] = 0.148038$$

$$\varepsilon_a = \left| \frac{0.148038 - 0.147605}{0.148038} \right| \times 100 = 0.293\%$$

it-2:-

$$f_y^{(2)} = \frac{-2(0.148028)}{4.5} = -0.065795$$

$$y_y^{(3)} = 0.1875 + \frac{0.5}{4}[9(-0.065795) - 1.303735] = 0.148002$$

$$\varepsilon_a = \left| \frac{0.148002 - 0.148038}{0.148002} \right| \times 100 = 0.024\%$$

it-4:-

$$f_y^{(3)} = \frac{-2(0.148002)}{4.5} = -0.065779$$

$$y_4^{(4)} = 0.1875 + \frac{0.5}{24}\left[9(-0.065779) - 1.303735\right]$$

$$= 0.148005$$

$$\varepsilon_a = \left| \frac{0.148005 - 0.148002}{0.148005} \right| \times 100 = 0.002\%.$$

$$\varepsilon_a = 0.002\% < \varepsilon_s .$$

$$\boxed{y(4.5) = 0.148}$$